CLASS 9 · CBSE · GANITA MANJARI PART 2 · SHORT NOTES

Chapter 12: Quadrilaterals

Topic-wise short notes with solved examples

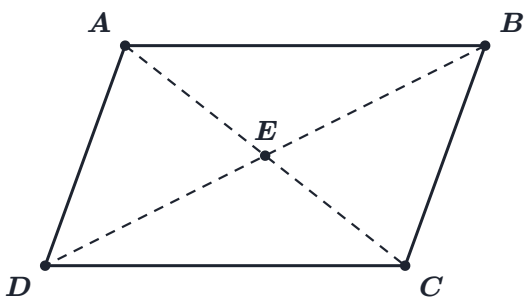
This chapter defines a quadrilateral carefully, proves the tests for a parallelogram, and uses parallelograms to prove the Midpoint Theorem, the Centroid Theorem and Varignon's theorem. It ends with tiling the plane using any quadrilateral.

1 What a quadrilateral is

- A **quadrilateral** (4-gon) $ABCD$ is formed by four points A, B, C, D in a plane, no three of them collinear, joined by the segments AB, BC, CD and DA, where every point other than the four vertices lies on exactly one of these segments.
- This rules out three collinear vertices (which would give a triangle), four points not in one plane (a non-planar figure) and sides that cross each other (a self-intersecting figure).
- **Sides**: AB, BC, CD, DA. **Diagonals**: AC and BD. **Adjacent** sides share a vertex and **opposite** sides do not. Angles at the ends of a side are adjacent angles.
- The same quadrilateral can be named $BCDA$, $CDAB$, $DABC$ or in the reverse order, but not $ACBD$.
- A quadrilateral is **convex** when all its interior angles are less than 180°. Then its diagonals intersect. A **non-convex** quadrilateral has one angle greater than 180° (a dent), and its diagonals do not intersect.
- The interior angles of any quadrilateral add up to 360°.

2 Parallelograms and their properties

- A **parallelogram** is a quadrilateral whose opposite sides are parallel. Rectangles, rhombuses and squares are parallelograms.
- **Theorem 1**: In a parallelogram (a) opposite sides are equal (b) opposite angles are equal (c) the diagonals bisect each other.



ABCD is a parallelogram. E is the mid-point of both diagonals.

- (a) The diagonal AC is a transversal for both pairs of parallel sides, so $\triangle ACD \cong \triangle CAB$ by ASA. Hence $AB = DC$ and $AD = BC$.
- (b) Adjacent angles are interior angles on the same side of a transversal, so they add up to 180°. It follows that $\angle A = \angle C$ and $\angle B = \angle D$.
- (c) $\triangle AED \cong \triangle CEB$ (using $AD = BC$ and the alternate angles), so $EA = EC$ and $EB = ED$.

Example 2.1 In a parallelogram $ABCD$, $\angle A = (3x + 10)^\circ$ and $\angle B = (2x + 20)^\circ$. Find all four angles.

SOLUTION

Adjacent angles are supplementary: $(3x + 10) + (2x + 20) = 180$, so $x = 30$.

$\angle A = \angle C = 100^\circ$ **and** $\angle B = \angle D = 80^\circ$

Example 2.2 The diagonals of a parallelogram $ABCD$ meet at E. If $AC = 16$ cm and $BD = 10$ cm, find AE and BE.

SOLUTION

The diagonals bisect each other.

$AE = 8$ **cm and** $BE = 5$ **cm**

3 Tests for a parallelogram

- The converses of the three properties are all true, and there is one more useful test.

Test	If a quadrilateral has … then it is a parallelogram
Theorem 2	both pairs of opposite sides equal
Theorem 3	both pairs of opposite angles equal
Theorem 4	diagonals that bisect each other
Theorem 5	one pair of opposite sides both equal and parallel

- Proof of Theorem 2: $\triangle ACD \cong \triangle CAB$ by SSS, so the alternate angles are equal and the opposite sides are parallel.
- Proof of Theorem 3: $2(\angle A + \angle B) = 360^\circ$, so $\angle A + \angle B = 180^\circ$ and $AD \parallel BC$. In the same way $AB \parallel DC$.
- Proof of Theorem 4: $\triangle AED \cong \triangle CEB$ by SAS, which gives equal alternate angles.
- Proof of Theorem 5: if $AB \parallel DC$ and $AB = DC$, then $\triangle EAB \cong \triangle ECD$ by ASA, so the diagonals bisect each other. Now use Theorem 4.
- Special parallelograms: a parallelogram with one right angle is a **rectangle**. A parallelogram with equal diagonals is a rectangle. A parallelogram whose diagonal bisects an angle is a **rhombus**.

Example 3.1 Show that $A(1,1)$, $B(5,2)$, $C(7,6)$ and $D(3,5)$ are the vertices of a parallelogram.

SOLUTION

Mid-point of $AC = \left(\frac{1+7}{2}, \frac{1+6}{2}\right) = (4, 3.5)$. Mid-point of $BD = \left(\frac{5+3}{2}, \frac{2+5}{2}\right) = (4, 3.5)$.

The diagonals bisect each other, so $ABCD$ is a parallelogram (Theorem 4).

Example 3.2 One pair of opposite sides of a quadrilateral are equal and the other pair are parallel. Must it be a parallelogram?

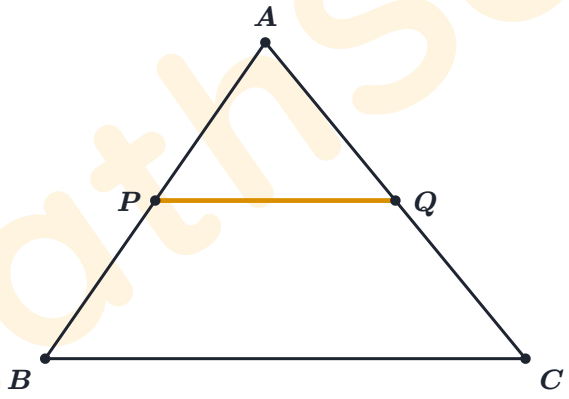
SOLUTION

Theorem 5 needs the **same** pair of sides to be equal and parallel.

No. An isosceles trapezium is a counterexample.

4 The Midpoint Theorem

- **Theorem 6 (Midpoint Theorem)**: The segment joining the mid-points of two sides of a triangle is parallel to the third side and is half as long.



P and Q are mid-points. PQ is parallel to BC and PQ = ½ BC.

- Proof idea: draw a line through C parallel to BA, meeting PQ produced at R. Then $\triangle APQ \cong \triangle CRQ$ (AAS), so $PQ = QR$ and $CR = AP = BP$. Since CR is equal and parallel to BP, $BCRP$ is a parallelogram (Theorem 5). Hence $PQ \parallel BC$ and $PQ = \frac{1}{2}PR = \frac{1}{2}BC$.
- **Theorem 7 (Converse)**: The line through the mid-point of one side of a triangle, parallel to another side, bisects the third side.
- Joining the mid-points of all three sides divides a triangle into **four congruent triangles**. Each has sides half as long as the original.

Example 4.1 The sides of a triangle are 10 cm, 14 cm and 18 cm. Find the perimeter of the triangle formed by joining the mid-points of its sides.

SOLUTION

Each side of the new triangle is half of one side: 5 cm, 7 cm and 9 cm.

21 cm

Example 4.2 In $\triangle ABC$, P is the mid-point of AB and the line through P parallel to BC meets AC at Q. If $AC = 13$ cm and $BC = 12$ cm, find AQ and PQ.

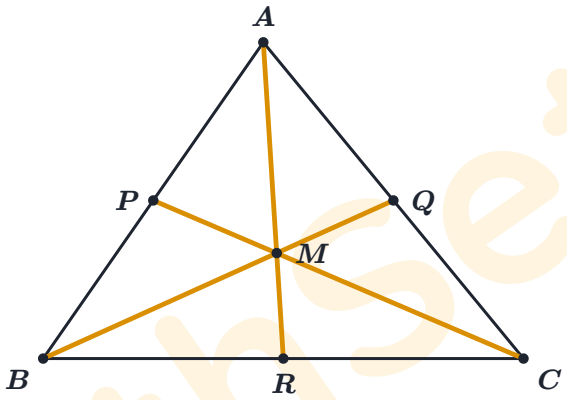
SOLUTION

By the converse of the Midpoint Theorem, Q is the mid-point of AC.

$AQ = 6.5$ cm and $PQ = 6$ cm

5 Medians and the centroid

- A **median** joins a vertex of a triangle to the mid-point of the opposite side.
- **Theorem 8 (Centroid Theorem)**: The three medians of a triangle pass through one point, the **centroid**. It divides each median in the ratio $2 : 1$, with the longer part towards the vertex.



M is the centroid. AM : MR = 2 : 1.

- Proof idea: let two medians CP and BQ meet at M, and let X, Y be the mid-points of BM and CM. By the Midpoint Theorem in $\triangle ABC$ and in $\triangle MBC$, both PQ and XY are parallel to BC and equal to half of it. So $\triangle MPQ \cong \triangle MYX$, which gives $MP = MY = YC$ and $MQ = MX = XB$. The same argument with another pair of medians gives the same point.

Example 5.1 A median of a triangle is 12 cm long. How far is the centroid from the vertex and from the mid-point of the opposite side?

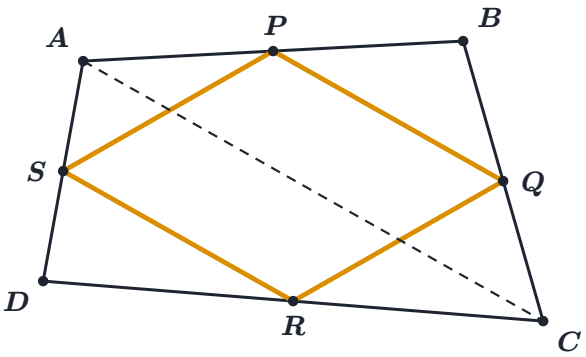
SOLUTION

The centroid divides the median in the ratio $2 : 1$: $\frac{2}{3} \times 12$ and $\frac{1}{3} \times 12$.

8 cm from the vertex and 4 cm from the mid-point

6 The Midpoint Theorem for quadrilaterals

- **Theorem 9 (Varignon's theorem)**: The mid-points of the four sides of any quadrilateral are the vertices of a parallelogram, called the **Varignon parallelogram**.



PQ and SR are both parallel to the diagonal AC

- Proof: by the Midpoint Theorem in $\triangle ABC$ and $\triangle ADC$, both PQ and SR are parallel to AC and equal to half of it. So PQ is equal and parallel to SR.
- The sides of the Varignon parallelogram are half the diagonals. So its perimeter equals the sum of the two diagonals of the quadrilateral.
- If the diagonals are **equal**, it is a rhombus. If they are **perpendicular**, it is a rectangle.

Example 6.1 The diagonals of a quadrilateral are 12 cm and 16 cm. Find the sides and the perimeter of its Varignon parallelogram.

SOLUTION

The sides are half the diagonals: 6 cm and 8 cm.

Sides 6 cm, 8 cm, 6 cm, 8 cm; perimeter 28 cm

7 Tiling the plane

- To **tile** the plane is to cover it with copies of a shape, leaving no gaps and no overlaps.
- Any parallelogram tiles the plane: a grid of two sets of parallel lines does it.
- **Any quadrilateral tiles the plane.** Its four angles add up to $360°$, so four copies fit around a point with each angle used once.
- Method: rotate a copy through $180°$ about the mid-point of a side. It fits exactly along that side. Repeat at every side of every copy.
- In such a tiling the Varignon parallelograms of the copies form a grid of parallelograms.
- A shape can tile around a point only if its angles can add up to $360°$ there. A regular pentagon cannot tile the plane: each angle is $108°$, and 360 is not a multiple of 108.

Example 7.1 Can a regular hexagon tile the plane?

SOLUTION

Each interior angle of a regular hexagon is $120°$, and $3 \times 120° = 360°$.

Yes. Three hexagons meet at each vertex, as in a honeycomb.

8 Common mistakes

- Using Theorem 5 with one pair of sides equal and the **other** pair parallel. The same pair must be both.
- Saying the diagonals of every parallelogram are equal or perpendicular. They only bisect each other.
- Writing the centroid ratio the wrong way round. The part from the vertex is the longer one.
- Assuming the converse of a property without proof. Each converse here has its own proof.
- Naming the vertices out of order: $ABCD$ means A is joined to B, B to C, C to D and D to A.

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